

Translations of tangent functions



1 B

2 Vertical translation by 4 units upwards.

3 True

4

a No

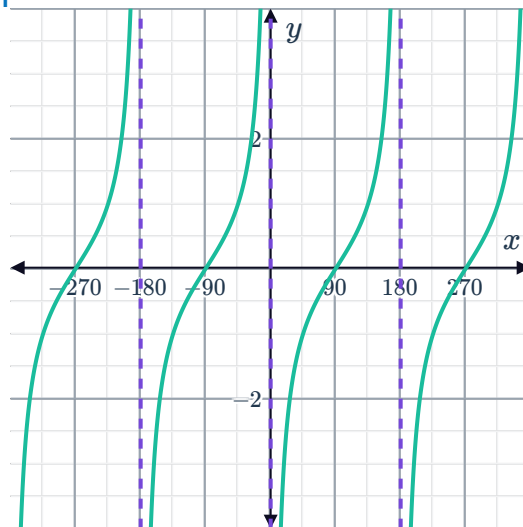
e 0

b 180°

c 180°

d 180°

f



5

a $-\sqrt{3}$

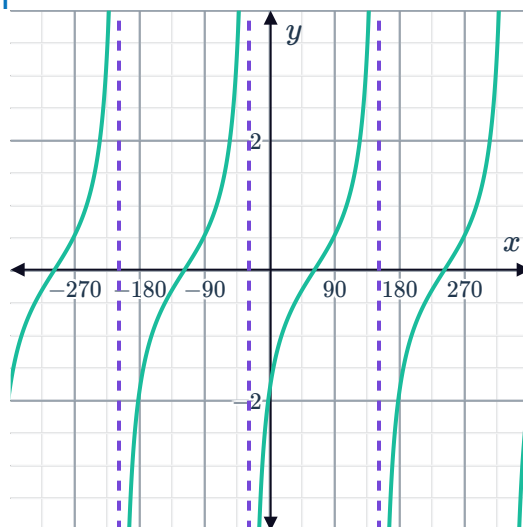
e $(-30)^\circ$

b 180°

c 180°

d 150°

f



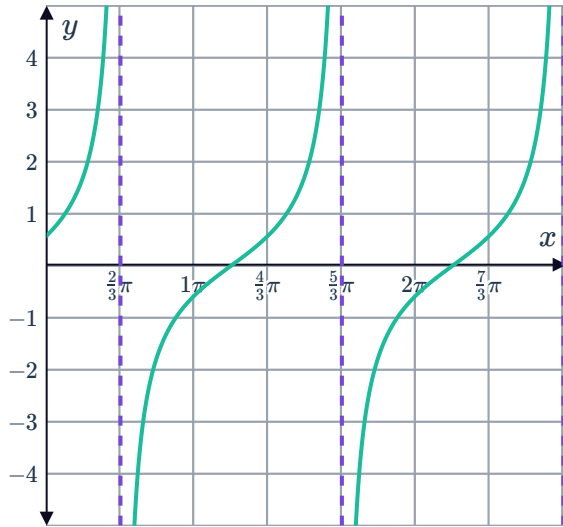
6

a $x = \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}$



b $x = \frac{7\pi}{6}, \frac{13\pi}{6}$

c

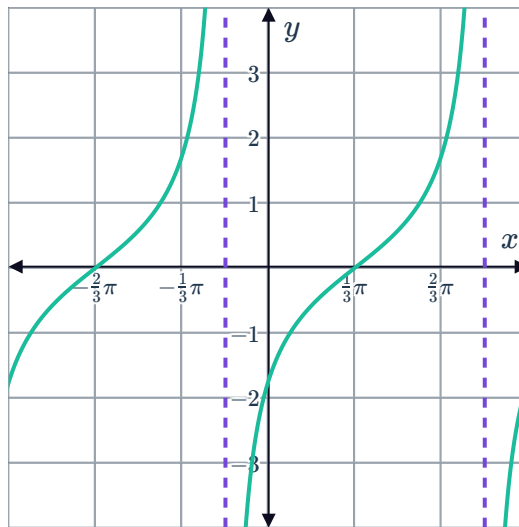


7

a i $\left(\frac{\pi}{3}, 0\right)$ ii $\left(\frac{7\pi}{12}, 1\right)$ iii $\left(\frac{5\pi}{6}, 0\right)$

b Horizontal translation (phase shift) of $\frac{\pi}{3}$ units right

c



8

a i Yes ii Yes iii Yes iv No

b A

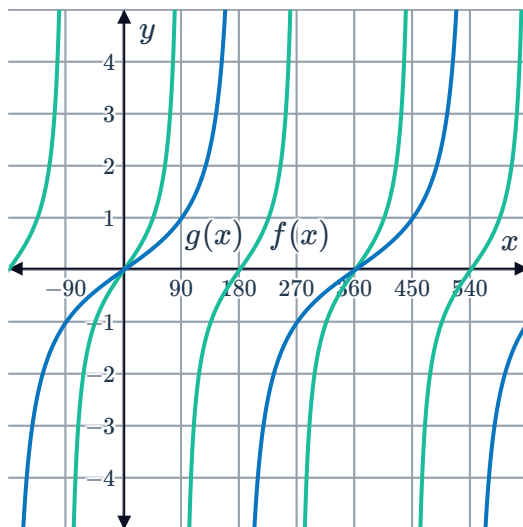
- 9 $f(x)$ and $g(x)$ have the same range, but different domains.



Subtracting 45° from x translates a function horizontally, which changes the position of the asymptote.

Dilations of tangent functions

10



11

a

Function	Period
$y = \tan x$	π
$y = \tan 2x$	$\frac{\pi}{2}$
$y = \tan 3x$	$\frac{\pi}{3}$
$y = \tan 4x$	$\frac{\pi}{4}$

b $\frac{\pi}{b}$

c The period becomes smaller.

12

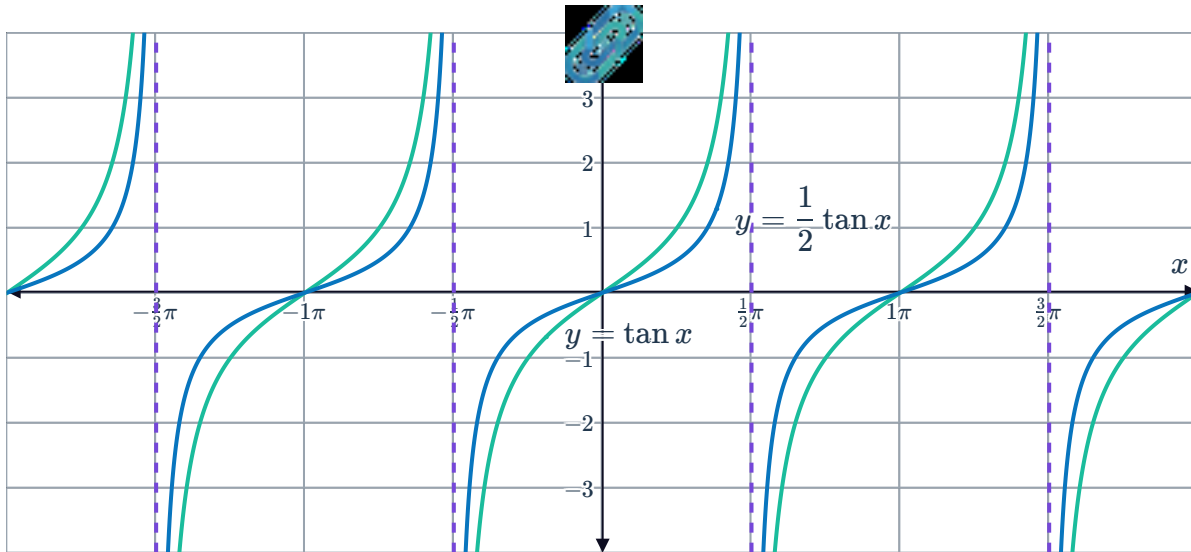
a 60°

b $\tan 3x$

13 $g(x) = \tan(4x)$

14 Vertical stretch by a factor of 3.

15



16 The graph of $y = \tan x$ has been stretched vertically by a factor of 5 and reflected across the x -axis.

17

a Reflecting the graph of $y = \tan x$ about the x -axis.

b $y = -\tan x$

18

a

i $y = 0$

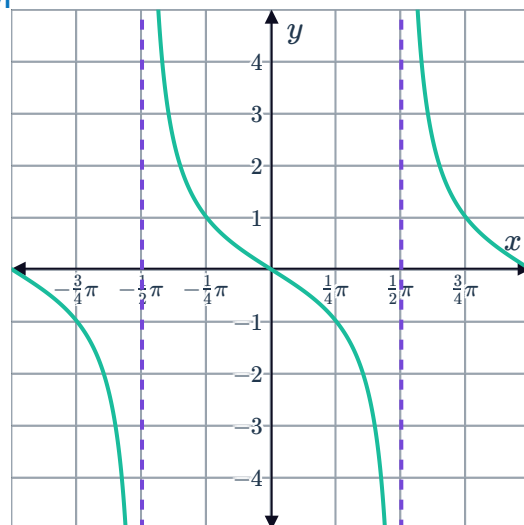
ii π

iii π

iv $x = \frac{\pi}{2}$

v $x = -\frac{\pi}{2}$

vi



b

i $y = \sqrt{3}$

v $x = -\frac{5\pi}{6}$

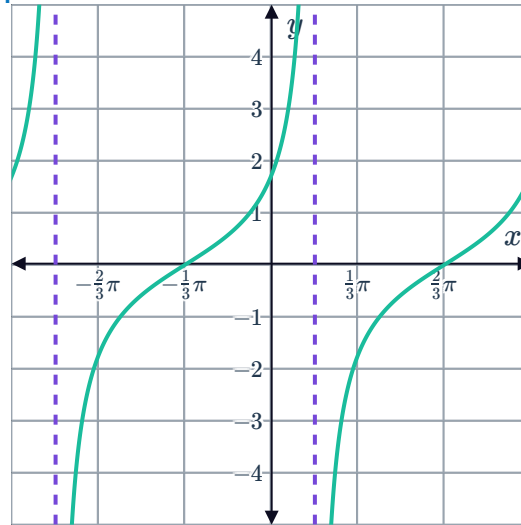
ii π



iii π

iv $x = \frac{\pi}{6}$

vi



c

i $y = 0$

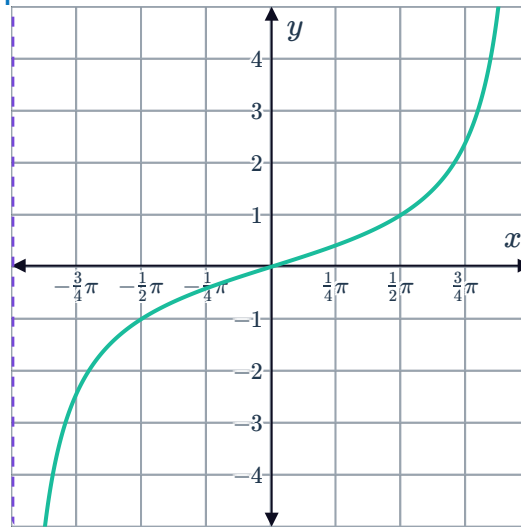
v $x = -\pi$

ii 2π

iii 2π

iv $x = \pi$

vi





Series of transformations

19

a

i $y = 8$

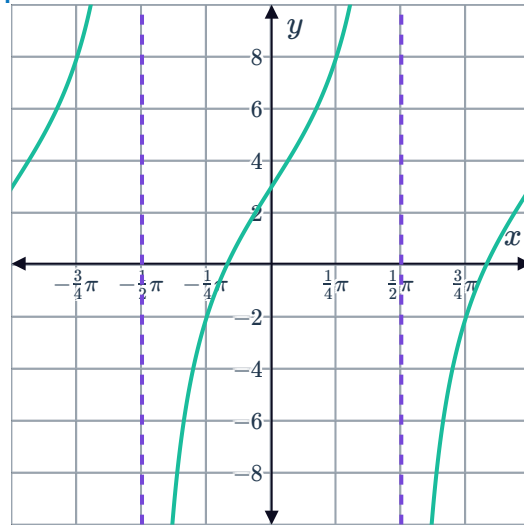
v $x = -\frac{\pi}{2}$

ii π

iii π

iv $x = \frac{\pi}{2}$

vi



b

i $y = -4$

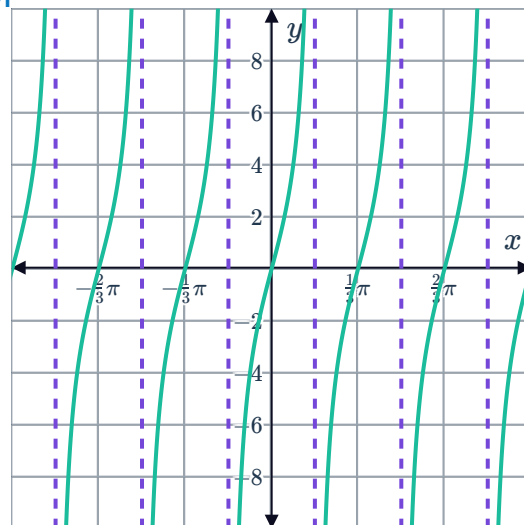
v $x = -\frac{\pi}{6}$

ii $\frac{\pi}{3}$

iii $\frac{\pi}{3}$

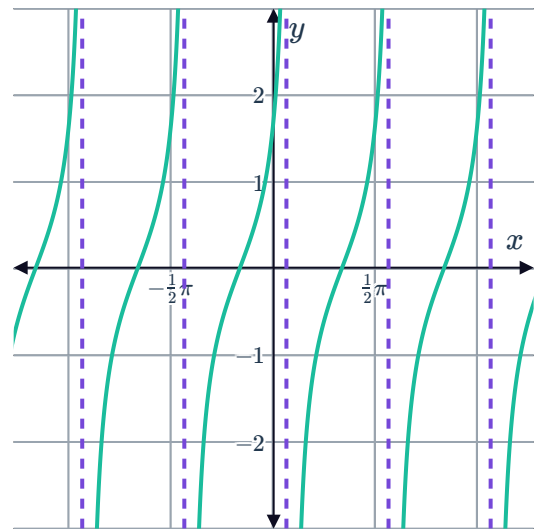
iv $x = \frac{\pi}{6}$

vi



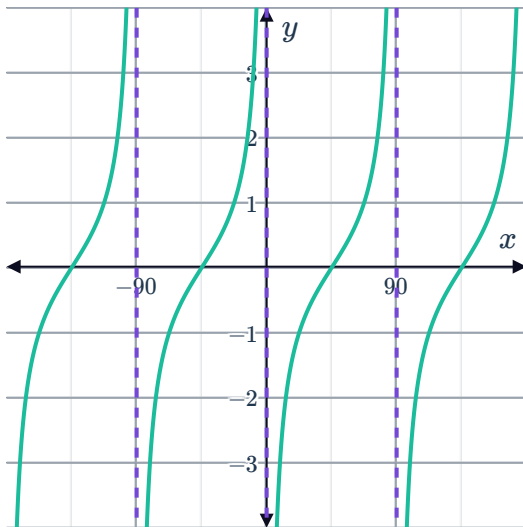
20

a $y = \tan \left(2 \left(x + \frac{\pi}{6} \right) \right)$

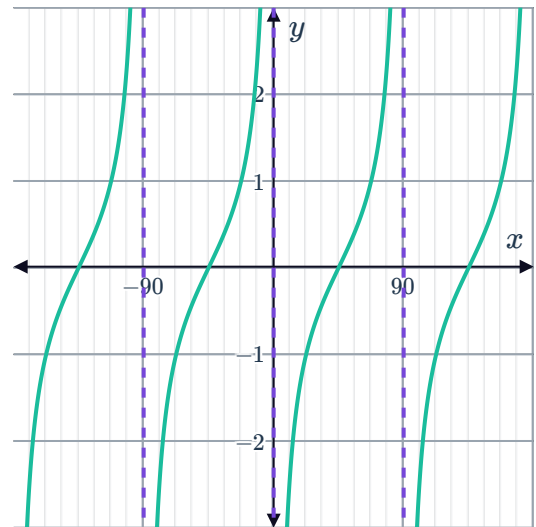


21

a



b



22

a No

b Yes

c No

d Yes

23 $y = \tan(-x) - 3$

24

a Compressing the graph of $y = \tan x$ horizontally, and translating it vertically.

b $y = \tan 4x + 4$

25



- a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k .
- b $(-\infty, \infty)$